

Fixed-Point Iteration Root Solver in MATLAB

Numerical Methods Portfolio Project

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Developed Using

MATLAB Online

Project Date

June 2026

Keywords: Numerical Methods, MATLAB, Fixed Point Iteration, Root Finding, Non-linear Equations, Convergence Analysis, Engineering Computation

TABLE OF CONTENTS

1	Project Overview.....	3
2	Objectives	3
3	Problem Statement.....	3
4	Theory	3
4.1	Convergence Condition.....	4
4.2	Divergence Demonstration	5
5	Nomenclature Table	5
6	Algorithm.....	6
7	Code Improvements	6
8	Flowchart	7
9	MATLAB Code	8
10	Output.....	9
11	Convergence Table	9
12	Graphical Output.....	10
12.1	Figure: Fixed Point Iteration Graph	10
13	Robustness Testing.....	11
13.1	Results	11
13.2	Discussion.....	11
14	Comparison with Other Methods.....	12
14.1	Discussion.....	12
14.2	Key Observation.....	13
15	Applications	11
16	Results.....	13
17	Discussion	13
18	Conclusion.....	14
19	Future Improvements	14

1 PROJECT OVERVIEW

The Fixed Point Iteration Method is a numerical technique used to find the roots of nonlinear equations by repeatedly evaluating a rearranged form of the original equation. Unlike several other root-finding methods, its success depends on how the equation is transformed into the form

$$x = g(x)$$

In this project, MATLAB is used to implement the Fixed Point Iteration Method for solving the nonlinear equation:

$$x^3 - x - 2 = 0$$

The project demonstrates the iterative procedure, convergence behavior, graphical interpretation, and the effect of different initial guesses on the solution. It also highlights an important concept in numerical analysis: a method may converge or diverge depending on the chosen iteration function.

2 OBJECTIVES

The objectives of this project are:

- Implement the Fixed Point Iteration Method in MATLAB.
- Determine the root of a nonlinear equation numerically.
- Study the convergence behavior of the iterative process.
- Investigate the effect of different initial guesses.
- Visualize the solution graphically using MATLAB plots.
- Demonstrate the convergence condition for Fixed Point Iteration.
- Compare the obtained root with those from other root-finding methods studied previously.

3 PROBLEM STATEMENT

The objective is to determine the root of the nonlinear equation:

$$x^3 - x - 2 = 0$$

using the Fixed Point Iteration Method. The equation is first rearranged into an equivalent form suitable for iteration:

$$x = \sqrt[3]{x + 2}$$

The iterative scheme is then applied repeatedly until the difference between successive approximations falls below a specified tolerance. The convergence characteristics and accuracy of the method are analyzed using MATLAB.

4 THEORY

The Fixed Point Iteration Method is an iterative numerical technique used to solve nonlinear equations. The method transforms a given equation into the form:

$$x = g(x)$$

A value is repeatedly substituted into the function $g(x)$ until successive approximations become sufficiently close to each other.

For this project, the nonlinear equation is:

$$x^3 - x - 2 = 0$$

Rearranging gives:

$$\begin{aligned} x^3 &= x + 2 \\ x &= \sqrt[3]{x + 2} \end{aligned}$$

Therefore, the iteration function becomes:

$$g(x) = \sqrt[3]{x + 2}$$

The Fixed Point Iteration formula is:

$$x_{n+1} = g(x_n)$$

Starting from an initial guess x_0 , the method generates a sequence of approximations until the desired accuracy is achieved.

4.1 CONVERGENCE CONDITION

A Fixed Point Iteration scheme converges only if the iteration function satisfies a specific condition near the root:

$$|g'(x)| < 1$$

If this condition is satisfied, successive approximations move closer to the root and the method converges.

For the selected iteration function:

$$g(x) = (x + 2)^{1/3}$$

the derivative is:

$$g'(x) = \frac{1}{3(x + 2)^{2/3}}$$

Near the root $x \approx 1.52138$, the magnitude of $g'(x)$ is less than 1, ensuring convergence.

If an alternative rearrangement produces $|g'(x)| > 1$ near the root, the iterations may diverge and move away from the solution.

This dependence on the choice of $g(x)$ is the distinguishing feature of the Fixed Point Iteration Method and highlights the importance of proper equation rearrangement in numerical analysis.

4.2 DIVERGENCE DEMONSTRATION

The convergence of the Fixed Point Iteration Method depends on the choice of the iteration function $g(x)$. A poor rearrangement of the original equation may cause the iterations to diverge instead of approaching the root.

Consider the alternative rearrangement:

$$g(x) = x^3 - 2$$

which gives the iteration formula:

$$x_{n+1} = x_n^3 - 2$$

Using this function, successive approximations move away from the root and rapidly increase in magnitude. The method fails to converge because the convergence condition

$$|g'(x)| < 1$$

is not satisfied near the root.

For this iteration function:

$$g'(x) = 3x^2$$

At the root $x \approx 1.52138$,

$$|g'(x)| \approx 6.94 > 1$$

Therefore, the iterations diverge.

This example demonstrates that Fixed Point Iteration is highly sensitive to the choice of $g(x)$. Unlike Bisection or False Position methods, convergence is not guaranteed unless an appropriate rearrangement is selected.

5 NOMENCLATURE TABLE

Symbol	Description
x	Unknown variable
$f(x)$	Original nonlinear function
$g(x)$	Iteration function
x_0	Initial guess
x_n	Approximation at iteration (n)
$x(n+1)$	Next approximation
Tol	Convergence tolerance
Iter	Iteration counter
$g'(x)$	Derivative of iteration function

6 ALGORITHM

The Fixed Point Iteration Method follows these steps:

1. Define the iteration function $g(x)$.
2. Choose an initial guess x_0 .
3. Specify a convergence tolerance.
4. Compute the next approximation using:

$$x_{n+1} = g(x_n)$$

5. Calculate the difference between successive approximations:

$$|x_{n+1} - x_n|$$

6. Check the convergence criterion:
 - If the difference is smaller than the specified tolerance, stop the iteration.
 - Otherwise, continue.
7. Replace the old approximation with the new value.
8. Repeat the process until convergence is achieved.
9. Display the final root and the number of iterations required.

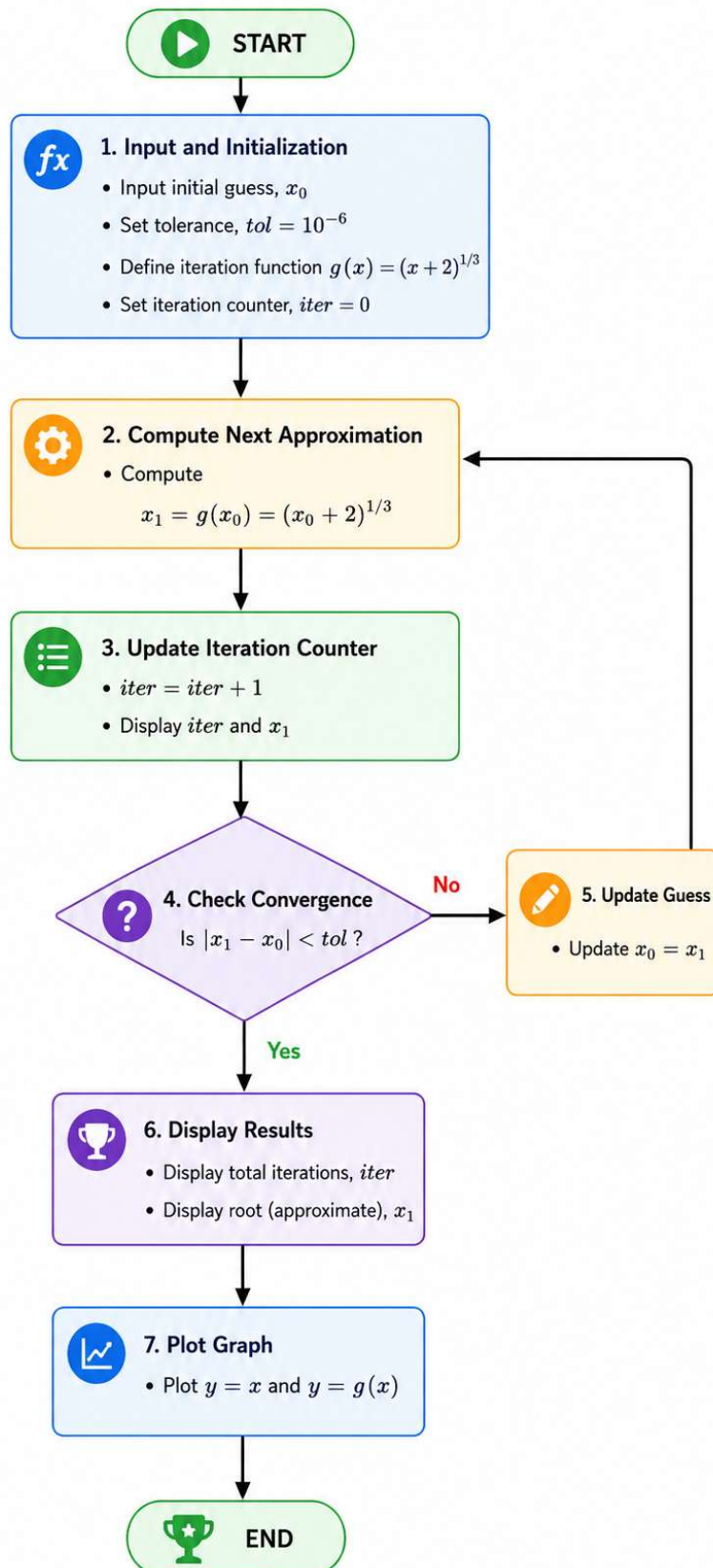
7 CODE IMPROVEMENTS

Several enhancements were incorporated into the MATLAB implementation to improve reliability and readability:

- Anonymous functions were used to define the iteration function efficiently.
- Iteration values were displayed in a formatted table for easier analysis.
- A convergence tolerance of 10^{-6} was implemented to ensure accurate results.
- Automatic iteration counting was included.
- Graphical visualization was added to illustrate the intersection of $y = x$ and $y = g(x)$.
- A maximum iteration limit can be added to prevent infinite loops in case of divergence.
- Multiple initial guesses were tested to evaluate the robustness of the method.
- A divergence study was performed using an alternative iteration function to demonstrate the importance of the convergence condition.

This implementation provides both numerical and graphical insight into the behavior of the Fixed Point Iteration Method.

Fixed Point Iteration Method Flowchart



The flowchart illustrates the logical sequence of the Fixed Point Iteration Method implemented in MATLAB. The process begins with initialization, repeatedly updates the approximation, checks convergence, and terminates when the specified tolerance is satisfied.

9 MATLAB CODE

```
% Rearranging
%  $x^3 - x - 2 = 0$ 
%  $x^3 = x + 2$ 
%  $x = (x + 2)^{(1/3)}$ 

% Define g(x)
g = @(x) (x + 2).^(1/3);

% Initial guess
x0 = input('Enter initial guess: ');

% Tolerance
tol = 1e-6;

iter = 0;

fprintf('Iteration\t x\n')

while true

    x1 = g(x0);

    iter = iter + 1;

    fprintf('%d\t\t %.8f\n', iter, x1)

    if abs(x1-x0) < tol
        break
    end

    x0 = x1;
end

fprintf('Iterations = %d\n',iter)
fprintf('Root = %.6f\n',x1)

% Graph
x = 0:0.01:3;

figure
plot(x, x, 'LineWidth', 2)
hold on
plot(x, g(x), 'LineWidth', 2)

grid on
legend('y = x', 'y = g(x)')
xlabel('x')
ylabel('y')
title('Fixed Point Iteration Method')
```


10 SAMPLE OUTPUT

For the initial guess:

$$x_0 = 1.5$$

MATLAB produced the following output:

```
Enter initial guess: 1.5
Iteration      x
1             1.51829449
2             1.52093526
3             1.52131570
4             1.52137049
5             1.52137838
6             1.52137952
7             1.52137968
Iterations = 7
Root = 1.521380
```

The method converged in **7 iterations** and produced the approximate root:

$$x \approx 1.521380$$

11 CONVERGENCE TABLE

Iteration	(x)
1	1.51829449
2	1.52093526
3	1.52131570
4	1.52137049
5	1.52137838
6	1.52137952
7	1.52137968

The convergence table shows that the approximate value of x becomes stable as the number of iterations increases. The difference between successive values decreases until the tolerance condition is satisfied.

Final root obtained:

$$x \approx 1.521380$$

12 GRAPHICAL OUTPUT

The graphical interpretation of the Fixed Point Iteration Method is based on plotting the functions:

$$y = x$$

and

$$y = g(x) = \sqrt[3]{x+2}$$

The point where the two curves intersect represents the fixed point of the iteration and corresponds to the root of the original equation.

The MATLAB-generated graph shows a clear intersection near:

$$x \approx 1.52138$$

which agrees with the numerical solution obtained from the iteration process.

12.1 FIGURE: FIXED POINT ITERATION GRAPH

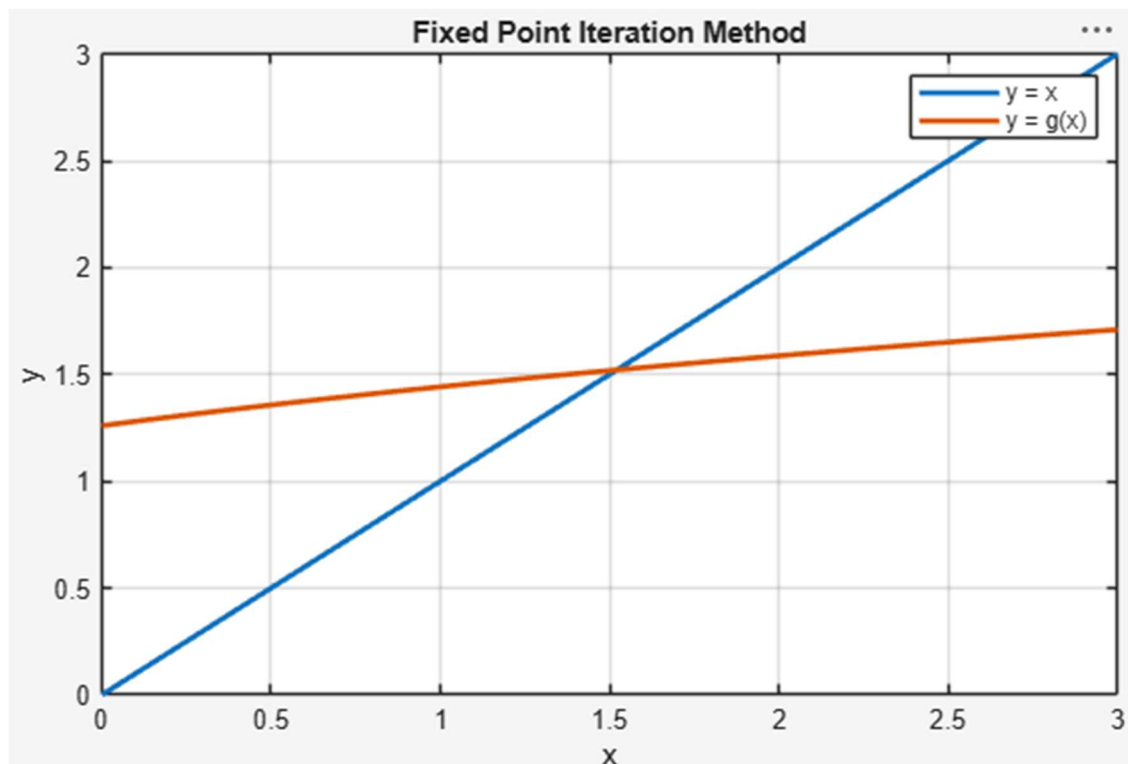


Figure 1. Graph of $y=x$ and $y=g(x)$. The intersection of the two curves represents the fixed point and the approximate root of the nonlinear equation.

13 ROBUSTNESS TESTING

To evaluate the stability of the Fixed Point Iteration Method, the algorithm was tested using different initial guesses. The objective was to determine whether the method converges to the same root regardless of the starting value.

The following initial guesses were used: $x_0 = 1$, $x_0 = 2$, and $x_0 = 5$.

13.1 RESULTS

Initial Guess	Root Obtained	Iterations
1	1.521380	8
2	1.521380	8
5	1.521380	9

13.2 DISCUSSION

The results show that the Fixed Point Iteration Method converged to the same root for all three initial guesses. Although the number of iterations varied slightly, the final solution remained unchanged.

The initial guesses of 1 and 2 required 8 iterations to satisfy the convergence criterion, while the initial guess of 5 required 9 iterations. This increase occurred because the starting value was farther from the actual root.

The successful convergence for multiple starting points indicates that the chosen iteration function

$$g(x) = \sqrt[3]{x + 2}$$

is stable and satisfies the convergence condition near the root. Therefore, the method demonstrates good robustness for this particular problem.

The experiment confirms that the Fixed Point Iteration Method can reliably determine the root of the equation when an appropriate iteration function is selected.

14 APPLICATIONS

The Fixed Point Iteration Method is widely used in engineering, mathematics, and scientific computing for solving nonlinear equations. Some common applications include:

- Root finding in engineering design problems.
- Solving nonlinear equations in thermodynamics and fluid mechanics.
- Iterative solutions of heat transfer and mass transfer equations.
- Numerical analysis and computational mathematics.
- Electrical circuit analysis involving nonlinear components.
- Structural and mechanical engineering calculations.
- Serving as the foundation for more advanced iterative numerical methods.

The method is particularly useful when a suitable rearrangement of the governing equation can be obtained and the convergence condition is satisfied.

15 COMPARISON WITH OTHER METHODS

The Fixed Point Iteration Method was compared with the other root-finding techniques implemented during this numerical methods study, namely the Bisection Method, False Position Method, Newton-Raphson Method, and Secant Method. All methods were applied to the same nonlinear equation:

$$x^3 - x - 2 = 0$$

and produced approximately the same root:

$$x \approx 1.521380$$

However, the methods differ in terms of convergence speed, required inputs, computational complexity, and reliability.

Method	Root Obtained	Iterations	Derivative Required	Initial Requirement	Characteristics
Bisection Method	1.521380	20	No	Interval $[a, b]$	Highly reliable but slow
False Position Method	1.521380	5	No	Interval $[a, b]$	Faster than Bisection
Newton-Raphson Method	1.521380	3	Yes	One initial guess	Fastest convergence
Secant Method	1.521380	10	No	Two initial guesses	Fast convergence without derivatives
Fixed Point Iteration	1.521380	7	No	One initial guess	Depends on choice of $g(x)$

15.1 DISCUSSION

The Bisection Method was the most robust technique because convergence is guaranteed whenever a sign-changing interval exists. However, it required the largest number of iterations.

The False Position Method improved upon Bisection by using linear interpolation instead of interval halving. As a result, it achieved the same root with significantly fewer iterations.

The Newton-Raphson Method exhibited the fastest convergence and reached the solution in only three iterations. Its major limitation is the requirement for derivative evaluation, which may be difficult for complex functions.

The Secant Method removed the need for derivatives while maintaining rapid convergence. It required two initial guesses and converged more quickly than bracketing methods.

The Fixed Point Iteration Method converged successfully in seven iterations and required only a single initial guess. Unlike the other methods, its convergence depended entirely on the selected iteration function $g(x)$. A suitable rearrangement led to convergence, whereas an unsuitable rearrangement caused divergence.

15.2 KEY OBSERVATION

The Fixed Point Iteration Method is unique among the studied methods because its success is governed by the convergence condition:

$$|g'(x)| < 1$$

near the root. This property makes the method an important educational tool for understanding convergence behavior in numerical analysis.

Overall, Newton-Raphson provided the fastest convergence, Bisection offered the highest reliability, False Position delivered an excellent balance between speed and stability, Secant eliminated derivative calculations while maintaining efficiency, and Fixed Point Iteration provided valuable insight into the role of convergence conditions in iterative numerical methods.

16 RESULTS

The Fixed Point Iteration Method successfully determined the root of the nonlinear equation:

$$x^3 - x - 2 = 0$$

The computed root was:

$$x = 1.521380$$

Using an initial guess of $x_0 = 1.5$, the method converged in 7 iterations.

Additional robustness testing using different initial guesses produced the same root, demonstrating the stability of the selected iteration function.

The graphical solution confirmed the numerical result through the intersection of $y = x$ and $y = g(x)$.

17 DISCUSSION

The Fixed Point Iteration Method successfully converged to the root of the nonlinear equation $x^3 - x - 2 = 0$. The method demonstrated a simple and intuitive approach in which the original equation was transformed into the form $x = g(x)$, and successive approximations were generated until convergence was achieved.

A significant observation from this study is that the convergence of the method depends strongly on the choice of the iteration function. The selected function,

$$g(x) = \sqrt[3]{x + 2}$$

satisfied the convergence condition near the root and produced stable results for multiple initial guesses.

Compared with the Bisection Method, Fixed Point Iteration required fewer iterations but lacked the guaranteed convergence provided by interval bracketing. Compared with Newton-Raphson, it converged more slowly but did not require derivative evaluation. Its performance was comparable to the Secant Method for this particular problem.

The robustness study showed that different starting values converged to the same root, indicating that the chosen iteration function was well-behaved in the region of interest. The graphical representation further reinforced the concept of a fixed point as the intersection of $y = x$ and $y = g(x)$.

The divergence demonstration highlighted an important limitation of the method. An unsuitable rearrangement can cause the iterations to move away from the root, making the method ineffective despite the existence of a valid solution.

The divergence experiment further highlighted the importance of the convergence condition. When the alternative iteration function $g(x) = x^3 - 2$ was used, the method failed to converge and the iterates moved away from the root. This behavior confirmed that Fixed Point Iteration is highly dependent on the mathematical form of the chosen iteration function.

18 CONCLUSION

The Fixed Point Iteration Method was successfully implemented in MATLAB to solve the nonlinear equation:

$$x^3 - x - 2 = 0$$

The method converged to the root:

$$x \approx 1.521380$$

with high accuracy and demonstrated stable performance for various initial guesses.

The study emphasized the importance of selecting an appropriate iteration function and understanding the convergence condition $|g'(x)| < 1$. Unlike several other root-finding techniques, the success of Fixed Point Iteration depends directly on the mathematical form of the rearranged equation.

Overall, the project provided valuable insight into iterative numerical methods, convergence analysis, and graphical interpretation of nonlinear equation solving.

19 FUTURE IMPROVEMENTS

Several enhancements can be incorporated into future versions of the project:

- Implement automatic convergence checking using the derivative condition $|g'(x)| < 1$.
- Add relative error calculations alongside absolute error.
- Include graphical visualization of iteration paths (cobweb diagrams).
- Allow the user to select different iteration functions and compare their performance.
- Introduce a maximum iteration limit to prevent infinite loops during divergence.
- Extend the implementation to solve systems of nonlinear equations.
- Develop a unified MATLAB application that compares multiple root-finding methods within a single interface.
- Integrate convergence plots showing error reduction versus iteration number.

These improvements would make the program more versatile and provide deeper insight into the behavior of iterative numerical methods.